EGDE INTELLIGENCE LAB ASSIGNMENT- 3/4/26

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LOG DATA ANALYTICS:

1. Objective

Design and implement a system that performs **real-time log data analytics at the edge** to reduce latency, bandwidth usage, and dependency on centralized cloud systems. The goal is to enable faster anomaly detection, monitoring, and decision-making.

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2. Background & Motivation

Traditional log analytics systems rely heavily on centralized cloud platforms, leading to:

- High latency in processing
- Increased network bandwidth usage
- Delayed anomaly detection

Edge computing addresses these challenges by **processing logs closer to the data source** (e.g., IoT devices, servers, microservices).

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3. System Architecture

Components:

1. **Log Sources**
    - Applications, microservices, IoT devices, servers
2. **Edge Layer (Core Focus)**
    - Lightweight log collectors (e.g., Fluent Bit)
    - Stream processing (filtering, aggregation)
    - Local anomaly detection using ML models
3. **Cloud Layer (Optional)**
    - Long-term storage
    - Batch analytics
    - Model training and updates

Data Flow:
Log Generation → Edge Processing → (Filtered Data) → Cloud Storage/Visualization

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4. Key Features

- **Real-time log filtering and aggregation at edge**
- **Anomaly detection using lightweight ML models**
- **Bandwidth optimization (send only relevant logs)**
- **Low-latency alerting system**
- **Scalable across distributed edge nodes**

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5. Technologies & Tools

- **Programming:** Python

- **Edge Processing:** Docker, Kubernetes (K3s for lightweight clusters)
- **Streaming:** Apache Kafka / MQTT
- **Log Collection:** Fluent Bit / Logstash
- **Analytics:** Apache Spark (edge or cloud), Pandas
- **Visualization:** Grafana, Kibana

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6. Methodology

1. Collect logs from distributed sources
2. Preprocess logs at edge (cleaning, parsing)
3. Apply filtering rules (remove noise)
4. Perform real-time analytics (counts, error rates)
5. Run anomaly detection (threshold-based or RL/ML models)
6. Send summarized insights to cloud
7. Visualize results and trigger alerts

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7. Use Cases

- Microservices monitoring
- IoT system health tracking
- Cybersecurity (intrusion detection)
- Server performance analysis

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8. Evaluation Metrics

- Latency reduction
- Bandwidth savings
- Detection accuracy
- Resource utilization at edge

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9. Expected Outcomes

- Faster insights with reduced delay
- Lower cloud dependency
- Efficient handling of high-volume logs
- Scalable and cost-effective analytics system

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10. Future Enhancements

- Integration with reinforcement learning for adaptive log filtering
- Federated learning across edge nodes
- Advanced predictive analytics